

Nordic Walking in Water on Cerebrovascular Reactivity and Cognitive Function in Elderly Patients with Type 2 Diabetes

TAWATCHAI PLOYDANG¹, WEERAPAN KHOVIDHUNKIT^{2,3}, HIROFUMI TANAKA⁴, and DAROONWAN SUKSOM^{1,5}

¹Faculty of Sports Science, Chulalongkorn University, Bangkok, THAILAND; ²Faculty of Medicine, Chulalongkorn University, Bangkok, THAILAND; ³Excellence Center in Diabetes, Hormone and Metabolism, King Chulalongkorn Memorial Hospital, Bangkok, THAILAND; ⁴Department of Kinesiology and Health Education, The University of Texas at Austin, Austin, TX; and ⁵Exercise Physiology in Special Population Research Unit, Chulalongkorn University, Bangkok, THAILAND

ABSTRACT

PLOYDANG, T., W. KHOVIDHUNKIT, H. TANAKA, and D. SUKSOM. Nordic Walking in Water on Cerebrovascular Reactivity and Cognitive Function in Elderly Patients with Type 2 Diabetes. *Med. Sci. Sports Exerc.*, Vol. 55, No. 10, pp. 1803–1811, 2023. **Introduction:** Aquatic Nordic walking (ANW) is a novel whole-body low-impact exercise that can be practiced by a variety of older adults with chronic conditions. However, its efficacy on several aspects of health is largely unknown. **Purpose:** This study aimed to determine the effects of regular ANW on glycemic control and vascular function in older adults with type 2 diabetes and mild cognitive impairment. **Methods:** Thirty-three older adults with type 2 diabetes age 60–75 yr were randomly allocated to nonexercising control ($n = 17$) or ANW ($n = 17$) groups. Nordic walking was performed in a pool at water temperature of 34°C–36°C, three times per week for 12 wk. **Results:** Measures of functional physical fitness including chair stand, timed up and go, chair sit and reach, reach and back scratch, and 6-min walk test scores were all improved after ANW (all $P < 0.05$). Plasma glucose, glycosylated hemoglobin, and homeostasis model assessment of insulin resistance decreased (all $P < 0.05$) in ANW. Vascular reactivity as assessed by brachial flow-mediated dilation increased, and arterial stiffness as assessed by brachial-ankle pulse wave velocity decreased in ANW (all $P < 0.05$). No significant changes were observed in the control group. Middle cerebral artery pulsatility index decreased with ANW under normocapnia condition ($P < 0.05$). Cerebrovascular conductance increased with ANW under hypercapnia condition. Montreal Cognitive Assessment score increased in the ANW group ($P < 0.001$). Changes in Montreal Cognitive Assessment scores were positively associated with corresponding changes in brain-derived neurotrophic factor ($r = 0.540$, $P = 0.031$). **Conclusions:** Nordic walking in water was a safe and effective innovative exercise modality to improve glycemic control, vascular function, physical fitness, cerebrovascular reactivity, and cognitive function in older adults with type 2 diabetes. **Key Words:** GLYCEMIC CONTROL, VASCULAR FUNCTION, HEALTH-RELATED PHYSICAL FITNESS

Type 2 diabetes mellitus is a complex metabolic disease that can lead to a variety of complications including cardiovascular disease and dementia (1). Both glycemic control and vascular dysfunction are implicated in the pathogenesis of both cognitive dysfunction and cardiovascular disease in diabetes (2). Most of the key measures of macrovascular and microvascular functions, including endothelium-dependent vasodilation, is depressed, and central artery stiffness is elevated in older adults with diabetes (3). A hypothesized physiological connection is that diabetes-induced vascular stiffening can lead

to increased transmission of pulsatile stress into the microcirculation and impair oxygen delivery to the brain (4). Because of the lack of energy reserve, the brain relies heavily on the perfusion through the vascular system to support its energy demand. In addition, excessive blood flow pulsatility is thought to cause cerebral microvascular damage, as high-flow and low-resistance organs like the brain would be highly susceptible to the damages induced by pulsatile stress (5). The impairment in vascular function can elicit the disturbance in the stringently regulated cerebral homeostasis and triggers a pathological cascade leading to cognitive decline. A strong correlation between aortic stiffness and middle cerebral pulsatility has been reported in patients with cerebrovascular disease (6). In adults with type 2 diabetes, the stiffening of cerebral vessels is associated with reduced cerebrovascular responsiveness to cognitive tasks and poorer cognitive performance (7).

Regular physical activity has been recommended for older patients with type 2 diabetes (8). However, the magnitude of benefits induced by regular exercise tends to be less than ideal on peripheral vascular and cerebrovascular functions (9). An innovative mode of exercise that exerts potentially large impacts

Address for correspondence: Daroonwan Suksom, Ph.D., Faculty of Sports Science, Chulalongkorn University, Rama 1 Rd, Pathumwan, Bangkok 10330, Thailand; E-mail: Daroonwan.S@chula.ac.th.

Submitted for publication December 2022.

Accepted for publication May 2023.

0195-9131/23/5510-1803/0

MEDICINE & SCIENCE IN SPORTS & EXERCISE®

Copyright © 2023 by the American College of Sports Medicine

DOI: 10.1249/MSS.0000000000003216

Healthcare) equipped with B-mode ultrasonography with a high-frequency linear-array transducer. The longitudinal ultrasound images were obtained over the common carotid artery between 1 and 2 cm proximal to the bifurcation. Arterial compliance was measured in the right common carotid artery using the combination of ultrasound imaging of a common carotid artery diameter and a noninvasive blood pressure (VP-1000plus; Omron Healthcare) (25). The ultrasound images were analyzed with the use of image analysis software (Carotid Analyzer; Medical Imaging Applications).

Flow-mediated dilation (FMD) was measured as an index of endothelium-dependent vasodilation by using the ultrasound machine (EPIQ 5 Ultrasound System; Philips Healthcare) equipped with a high-resolution linear-array transducer. The brachial artery of a dominant arm was imaged 5 cm above the antecubital fossa. The cuff set placed on the forearm was inflated rapidly to 50 mm Hg above systolic blood pressure for 5 min and deflated during recovery as previously described (26). Artery diameters were analyzed by ultrasound images via digital viewing software (Brachial Analyzer; Medical Imaging Applications). FMD was calculated using the formula: (peak postocclusion diameter – baseline diameter)/(baseline diameter) $\times$ 100.

Blood velocity in the MCA was measured noninvasively using transcranial Doppler ultrasonography (EPIQ 5 Ultrasound System; Philips Healthcare). A 1.8-MHz transcranial Doppler probe was attached to the patient's head at the right temporal window (27) in a supine position with the head of bed slightly elevated (30%–45%). After identifying the MCA, pulsed-wave Doppler is used to measure the velocity of blood flow within the vessel continuously. The Doppler sample gate is centered over the MCA's red color flow signal to obtain a spectral Doppler waveform. The difference between the long axis of the MCA itself and the angle of insonation was minimized (to <10–15°) for all participants. After the signals were identified at a depth of 45–60 mm, the probe was fixed with a probe folder so as to not change the insonating angle (28).

MCA blood velocity was measured separately in three different conditions; normocapnia, hypocapnia, and hypercapnia conditions as previously described (29). The initial partial pressure of carbon dioxide (PCO_2) was monitored as hypocarbic/hyperventilated ($PCO_2 < 35$ mm Hg), normocarbic ($PCO_2 35$ –45 mm Hg), and hypercarbic/hypoventilated ($PCO_2 > 45$ mm Hg). Hypocapnic cerebral vasoconstriction immediately before hypercapnic stimuli may influence ensuing vasodilation. Ventilatory gases were continuously measured and acquired via breath-by-breath using a flow sensor.

After the participant had rested in the supine position for at least 10 min, a nose clip was placed and the participant breathed through a mouthpiece with a Y-valve, with one end open to room air and the other end connected to a 5-L rebreathing bag. Baseline MCA blood velocity, heart rate, blood pressure, and end-tidal CO_2 ($EtCO_2$) were recorded simultaneously for 3 min. During these measurements, participants were instructed to breathe normally and to avoid body movement or Valsalva-like maneuvers. After baseline data collection, participants were

instructed to perform voluntary hyperventilation for 20 s (1 breath per second) to induce a brief period of hypocapnia. Heart rate and blood pressure (CARESCAPE V100; Dinamap; GE Healthcare) were obtained during the first 10 s of hyperventilation, whereas cerebral blood flow in MCA was recorded during the last 20 s of hyperventilation. After hyperventilation, a 5-min recovery period was provided to allow cerebral hemodynamics to return to the baseline. After $EtCO_2$ and MCA blood velocity returned to the baseline after hyperventilation, a respiratory valve was switched to an air reservoir containing 5% CO_2 and 21% O_2 , and the participants were asked to breathe spontaneously for 3 min. The air reservoir was continuously filled from a cylinder whose air pressure was manually adjusted to subjects' respiratory volume.

Then participants breathed air containing a gas mixture of 5% CO_2 and 21% O_2 balanced with nitrogen spontaneously for 3 min to induce hypercapnia. Briefly, at the end of a deep inspiration, the Y-valve of the mouthpiece was switched to the rebreathing bag for 3 min to induce a progressive increase in arterial CO_2 and then followed by a recovery period for 4 min. During rebreathing, a small amount of oxygen was added to the rebreathing bag based on each subject's basal metabolic rate to maintain constant arterial blood oxygen saturation. The data were recorded during the last minute of hypercapnia.

Pulsatility index (PI) was calculated as the difference between systolic and diastolic velocity divided by mean velocity (27,30). CVR was defined as a percent change in the MCA velocity (MCAv) over an absolute change in $EtCO_2$ (27). The change in CVR was estimated from the three different conditions of $EtCO_2$ levels including normocapnia to hypocapnia, normocapnia to hypercapnia, and hypocapnia to hypercapnia. Moreover, cerebrovascular conductance index (CVCi) was computed to account for the effects of changes in mean arterial pressure on MCA blood velocity during three conditions (27). CVR values were averages of >5 recorded values in each participant. The formulas used to determine CVR and CVCi are shown hereinafter:

$$CVR (\% \cdot \text{mm Hg}^{-1}) = \% \Delta \text{MCAv} / \Delta \text{End-tidal } CO_2$$

$$CVCi ((\text{cm} \cdot \text{s}^{-1}) \cdot \text{mm Hg}^{-1}) = \text{MCAv} / \text{MAP}$$

The MoCA evaluates multiple cognitive domains including visuospatial/executive function, naming, memory, attention, language, abstraction, delayed recall, and orientation. The MoCA was used also for cognitive screening to detect MCI between 18 and 24 scores from 30 scores (21), as MoCA is highly sensitive and effective in detecting MCI in patients with type 2 diabetes (31). Concurrent sensitivity and specificity of MoCA for identifying MCI patients in Thailand are 70% and 95% with high internal consistency (Cronbach $\alpha = 0.744$) (21). The Mini-Mental State Examination (MMSE) is consisted of seven domains including orientation to time, orientation to place, registration, attention and calculation, word recall, language, and visual construction. The Trail Making Test Part B was used to measure the executive function. Stroop color and word

test was used to assess the ability to inhibit cognitive interference (32).

After a 12-h overnight fasting, a venous blood sample was collected from the antecubital vein. Complete blood count (via automated cell counter), concentrations of fasting plasma glucose (enzymatic assay using hexokinase reaction), insulin (chemiluminescent microparticle immunoassay), high sensitivity C-reactive protein (immunoturbidimetric assay), lipids and lipoproteins (enzymatic assays), and HbA_{1c} (enzymatic assay) were measured in the certified clinical laboratory within the Faculty of Allied Health Science at Chulalongkorn University. Malondialdehyde concentration was estimated by thiobarbituric acid assay (Cayman, Ann Arbor, MI). Nitric oxide (NO) concentration was measured in a standard Griess reagents (Promega, Madison, MI). Tumor necrosis factor α , adiponectin, interleukin 1 β (IL-1 β), interleukin 6 (IL-6), and brain-derived neurotrophic factor (BDNF) were analyzed with the commercially available enzyme-linked immunosorbent assay kits (Sigma-Merck, Saint Louis, MO). Homeostasis model assessment of insulin resistance (HOMA-IR) was calculated using the following formula (33): [Fasting serum insulin (μ U·mL⁻¹) $\times$ Fasting plasma glucose (mg·dL⁻¹)]/405.

Statistical analyses. Sample size calculation was conducted using G*Power version 3.1.9.2 data analysis software (Franz Faul, Institute of Psychology, Kiel University, Germany) with the methods of ANOVA with repeated measures, within-between interaction, with power = 0.8, α level = 0.05, and effect size = 0.267. This effect size was from a previous study that examined the effect of aerobic training on cerebral blood flow in elderly with chronic kidney disease (34).

Descriptive data are expressed as means $\pm$ SD. All data were analyzed using SPSS (version 23; IBM, Armonk, NY). Grubb's outlier test (35) was used to detect outliers in the data sets using GraphPad statistical software (San Diego, CA). As a result, 1–3 outlier points were eliminated from some variables before the statistical analyses. Inclusion/elimination of these data points did not alter the primary findings. Before the parametric analyses, the normal distribution was confirmed using a Shapiro-Wilk test. A 2 $\times$ 2 (group–time) repeated-measures ANOVA, followed

by a *post hoc* test with LSD multiple comparison, were used to evaluate changes in all variables. Pearson's correlation coefficient was utilized to evaluate correlations between the percent pretest and posttest changes in BDNF, CVR normocapnia to hypocapnia condition, CVR hypocapnia to hypercapnia condition, CVCi hypercapnia condition, and changes in MoCA scores.

RESULTS

All the physical characteristics and blood pressure at baseline were similar between the groups (Table 1). The participants in ANW completed 100% of the training sessions (all 36 exercise sessions). Body weight, body mass index (BMI), fat mass, and lean mass did not change in both groups. Body fat percentage ($P = 0.009$) and heart rate at rest ($P = 0.010$) decreased significantly in ANW. Brachial blood pressure did not change significantly in both groups.

As shown in Table 2, flexibility as assessed by chair sit and reach and reach and back scratch improved with ANW ($P = 0.001$, $P = 0.027$), whereas no such improvements were seen in the control group. Muscle strength as estimated by chair stand and arm curl increased in ANW ($P = 0.001$, $P = 0.001$). Timed up and go test and 6-min walking performance improved in ANW ($P = 0.001$), whereas it decreased in the control group ($P = 0.001$). Two-minute step test improved only in ANW.

As shown in Table 3, carotid IMT and baPWV decreased in ANW ($P = 0.001$, $P = 0.015$). In addition, brachial FMD and arterial compliance increased in ANW ($P = 0.012$, $P = 0.010$) and higher than control ($P = 0.005$, $P = 0.001$). No such changes were observed in the control group.

As illustrated in Table 4, MCA blood velocity was not different between the groups at baseline and did not change with either intervention. PI decreased in ANW (pre vs post, 1.25 ± 0.15 vs 1.16 ± 0.10 ; $P = 0.003$) under normocapnia conditions but did not change significantly under hypocapnia or hypercapnia conditions. CVCi increased with ANW under hypercapnia conditions (pre vs post, 0.42 ± 0.90 vs 0.47 ± 0.13 (cm·s⁻¹)·mm Hg⁻¹; $P = 0.006$). There were no significant changes in CVR in both groups under normocapnia to

TABLE 1. Selected characteristics before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Variables	Nonexercising Control		ANW		ANOVA Statistics			
	Pre	Post	Pre	Post	Time	Group	Interaction	Effect Size
Male/female (n)	7/10	—	5/11	—				
Age (yr)	69.2 $\pm$ 5.3	—	68.9 $\pm$ 3.7	—				
Height (cm)	161 $\pm$ 8	—	159 $\pm$ 6	—				
Body weight (kg)	62.9 $\pm$ 9.3	62.5 $\pm$ 10.2	61.4 $\pm$ 10.1	60.5 $\pm$ 9.1	0.033	0.621	0.427	0.021
BMI (kg·m ⁻²)	24.8 $\pm$ 2.7	24.6 $\pm$ 3.1	24.2 $\pm$ 3.6	23.9 $\pm$ 3.3	0.041	0.561	0.330	0.031
Fat mass (kg)	20.9 $\pm$ 5.9	20.6 $\pm$ 6.0	21.8 $\pm$ 7.0	20.9 $\pm$ 6.2	0.014	0.768	0.176	0.058
Body fat (%)	34.0 $\pm$ 8.2	34.0 $\pm$ 7.8	35.8 $\pm$ 7.8	34.6 $\pm$ 7.4*	0.009	0.655	0.040	0.129
Lean mass (kg)	40.8 $\pm$ 8.2	40.9 $\pm$ 8.2	37.6 $\pm$ 6.2	37.6 $\pm$ 5.9	0.003	0.612	0.593	0.009
Bone mass (kg)	1.10 $\pm$ 0.19	1.09 $\pm$ 0.18	1.07 $\pm$ 0.11	1.08 $\pm$ 0.12	0.527	0.744	0.057	0.133
Waist circumference (cm)	86.7 $\pm$ 7.2	87.6 $\pm$ 8.1	86.7 $\pm$ 8.8	86.7 $\pm$ 9.3	0.351	0.856	0.351	0.028
Heart rate at rest (bpm)	72 $\pm$ 9	73 $\pm$ 8	75 $\pm$ 14	70 $\pm$ 10*	0.010	0.957	0.023	0.172
Systolic BP (mm Hg)	130 $\pm$ 16	125 $\pm$ 18	128 $\pm$ 11	122 $\pm$ 11	0.003	0.612	0.593	0.009
Diastolic BP (mm Hg)	72 $\pm$ 8	69 $\pm$ 9	73 $\pm$ 8	69 $\pm$ 8	0.002	0.893	0.694	0.005
Mean BP (mm Hg)	92 $\pm$ 10	87 $\pm$ 11	92 $\pm$ 8	86 $\pm$ 9	0.001	0.845	0.643	0.007

Values are means $\pm$ SD.

* $P < 0.05$ vs pre.

BP, blood pressure.

TABLE 2. Measures of physical fitness before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Physical Fitness	Nonexercising Control		ANW		ANOVA Statistics			
	Pre	Post	Pre	Post	Time	Group	Interaction	Effect Size
Chair sit and reach (cm)	5.2 ± 7.8	4.5 ± 8.9	1.9 ± 8.8	6.4 ± 9.5*	0.001	0.811	0.004	0.236
Reach and back scratch (cm)	-6.3 ± 15.8	-7.7 ± 14.7	-9.7 ± 10.9	-6.3 ± 10.6*	0.027	0.815	0.025	0.152
Chair stand 30 s (times)	15.5 ± 3.7	15.7 ± 4.3	15.3 ± 2.6	19.6 ± 3.2*,**	0.001	0.006	0.001	0.370
Arm curl 30 s (times)	15.9 ± 3.1	16.6 ± 4.0	17.7 ± 2.2	20.6 ± 2.5*,**	0.001	0.002	0.049	0.124
2-min step test (times)	168 ± 25	174 ± 29	171 ± 26	193 ± 23	0.009	0.171	0.109	0.083
Timed up and go test (s)	7.25 ± 1.37	7.95 ± 1.46*	7.39 ± 0.9	6.80 ± 0.82*,**	0.001	0.009	0.001	0.573
6-min walking (m)	496 ± 96	471 ± 88*	452 ± 61	492 ± 73*	0.001	0.677	0.001	0.428

Values are means ± SD.

* $P < 0.05$ vs pre.

** $P < 0.05$ vs control.

hypercapnia and hypocapnia to hypercapnia conditions. $\Delta\text{CVCi}/\Delta\text{EtCO}_2$ did not change significantly in either group.

There were no significant changes in hematological variables in any of the groups (Table 5). Glycemic control indicators including plasma glucose concentration, HbA_{1c} , and HOMA-IR decreased in ANW ($P = 0.033$, $P = 0.001$, $P = 0.044$) and lower than control ($P = 0.005$, $P = 0.020$, $P = 0.015$). Plasma lipid and lipoprotein concentrations and inflammatory markers did not change in either group. The concentration of NO and BDNF increased in ANW ($P = 0.002$, $P = 0.028$) and higher than control ($P = 0.030$, $P = 0.023$).

As illustrated in Table 6, MoCA score increased in the ANW group ($P = 0.001$). Notable changes were observed in domains of executive, abstraction, delayed recall, and orientation. MMSE score did not change significantly in both groups. There were no significant changes in the Stroop color test and Trail Making Test Part B in both groups.

The associations between cerebrovascular function and cognitive function measures of pre-post percent differences in the ANW group are shown in Figure 1. Changes in MoCA scores were positively associated with corresponding changes in BDNF ($r = 0.540$, $P = 0.031$), CVR normocapnia to hypocapnia condition ($r = 0.741$, $P = 0.001$), CVR hypocapnia to hypercapnia condition ($r = 0.509$, $P = 0.044$), and CVCi hypercapnia condition ($r = 0.550$, $P = 0.027$). There were no significant associations between HbA_{1c} and cerebral vascular measures.

DISCUSSION

Three months of moderate-intensity ANW produced favorable changes in multitudes of key measures in elderly patients with type 2 diabetes and MCI. Markers of glycemic control, indices of health-related physical fitness, and key subclinical measures of vascular dysfunction were all improved significantly with ANW. In addition, selected measures of CVR improved

with ANW, and these improvements were associated with corresponding improvements in cognitive function and BDNF. Taken together, these results indicate that Nordic walking performed in water is a highly effective innovative exercise in improving glycemic control, physical fitness, vascular function, CVR, and cognitive function in older adults with type 2 diabetes and MCI.

Twelve weeks of Nordic walking exercise on land produced significant reductions in body weight, BMI, HbA_{1c} (36), muscular flexibility, and aerobic fitness (37) in patients with type 2 diabetes. We have previously demonstrated that an addition of water immersion augmented the exercise training benefits of moderate cycling training in patients with type 2 diabetes (15). Because exercise is performed in water, viscosity of water results in a drag resistance during movements of arms and legs resulting in greater exertion under water (38) leading to increased metabolic rate and hence higher caloric expenditure during exercise (10). Accordingly, we hypothesized that Nordic walking performed in water could induce multitudes of benefits with great magnitudes. In addition to a decrease in body fat percentage, ANW examined in the present study produced significant reductions in fasting plasma glucose, HbA_{1c} , and insulin resistance as estimated by the HOMA-IR. We also observed significant increases in flexibility, both upper and lower muscular strength, and cardiovascular fitness in elderly patients with type 2 diabetes.

ANW improved endothelium-dependent vasodilation as assessed by brachial FMD, indices of arterial stiffness as assessed by baPWV, and carotid arterial compliance and reduced a marker of subclinical atherosclerosis, carotid artery IMT. These functional and structural changes in the vasculature are supported by circulating humoral markers, as a significant increase in plasma NO concentration was observed in the present study. These collective results indicate that water-based Nordic walking exercise is effective in improving a variety of macrovascular

TABLE 3. Vascular function and structure before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Vascular Measures	Nonexercising Control		Aquatic Nordic Walking		ANOVA Statistics			
	Pre	Post	Pre	Post	Time	Group	Interaction	Effect Size
Carotid IMT (mm)	0.68 ± 0.13	0.67 ± 0.13	0.68 ± 0.16	0.62 ± 0.13*	0.001	0.606	0.025	0.151
baPWV ($\text{cm}\cdot\text{s}^{-1}$)	1721 ± 302	1747 ± 205	1802 ± 225	1656 ± 176*	0.015	0.950	0.039	0.143
Arterial compliance (units)	0.24 ± 0.16	0.26 ± 0.11	0.30 ± 0.12	0.48 ± 0.17*,**	0.010	0.001	0.029	0.164
Brachial FMD (%)	4.8 ± 2.9	4.5 ± 2.7	4.9 ± 2.2	7.3 ± 2.7*,**	0.012	0.005	0.037	0.133

Values are means ± SD.

* $P < 0.05$ vs pre.

** $P < 0.05$ vs control.

TABLE 4. Cerebrovascular function before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Cerebrovascular Function	Nonexercising Control		ANW		ANOVA Statistics			Effect Size
	Pre	Post	Pre	Post	Time	Group	Interaction	
Mean CBFV (cm·s ⁻¹)								
Normocapnia	36.3 ± 8.7	34.3 ± 7.5	39.6 ± 12.8	40.2 ± 6.8	0.724	0.180	0.534	0.013
Hypocapnia	28.4 ± 7.2	27.6 ± 6.9	33.4 ± 13.0	30.6 ± 8.1	0.368	0.136	0.599	0.010
Hypercapnia	37.9 ± 7.9	37.3 ± 6.8	40.7 ± 11.5	43.0 ± 10.8	0.602	0.162	0.374	0.027
EtCO ₂ (mm Hg)								
Normocapnia	40.5 ± 11.5	33.7 ± 9.8*	38.8 ± 8.2	39.9 ± 5.5	0.008	0.404	0.028	0.146
Hypocapnia	31.0 ± 10.7	25.2 ± 8.2*	29.5 ± 6.5	30.1 ± 5.1	0.006	0.478	0.028	0.146
Hypercapnia	56.4 ± 9.5	46.9 ± 8.9*	51.0 ± 7.9	49.9 ± 4.6	0.001	0.629	0.002	0.263
MCA PI								
Normocapnia	1.33 ± 0.20	1.38 ± 0.22	1.25 ± 0.15	1.16 ± 0.10*,**	0.009	0.001	0.003	0.255
Hypocapnia	2.06 ± 0.51	2.08 ± 0.44	1.91 ± 0.30	1.78 ± 0.17	0.604	0.022	0.482	0.017
Hypercapnia	1.29 ± 0.20	1.27 ± 0.21	1.18 ± 0.19	1.09 ± 0.15	0.105	0.016	0.286	0.037
CVCI ((cm·s ⁻¹ ·mm Hg ⁻¹))								
Normocapnia	0.39 ± 0.08	0.38 ± 0.08	0.40 ± 0.10	0.43 ± 0.08	0.542	0.297	0.193	0.058
Hypocapnia	0.32 ± 0.10	0.30 ± 0.08	0.35 ± 0.11	0.35 ± 0.08	0.621	0.204	0.597	0.009
Hypercapnia	0.38 ± 0.80	0.35 ± 0.09	0.42 ± 0.09	0.47 ± 0.13*,**	0.007	0.005	0.006	0.223
CVR (%·mm Hg ⁻¹)								
Normocapnia-hypocapnia	2.7 ± 2.07	2.11 ± 1.56	3.33 ± 2.24	3.66 ± 1.48	0.788	0.032	0.350	0.031
Normocapnia-hypercapnia	0.36 ± 1.03	0.65 ± 1.04	1.09 ± 1.27	1.25 ± 1.91	0.549	0.056	0.863	0.001
Hypocapnia-hypercapnia	1.7 ± 1.19	1.85 ± 1.39	1.62 ± 1.42	2.56 ± 1.74	0.099	0.436	0.224	0.049
ΔCVCI/ΔEtCO ₂ (%·mm Hg ⁻¹)								
Normocapnia-hypocapnia	1.82 ± 2.34	1.76 ± 3.34	1.82 ± 2.06	3.41 ± 1.32	0.223	0.155	0.188	0.055
Normocapnia-hypercapnia	0.64 ± 1.64	0.76 ± 1.38	1.13 ± 1.97	1.15 ± 1.77	0.886	0.217	0.919	0.001
Hypocapnia-hypercapnia	1.7 ± 1.19	1.85 ± 1.39	1.62 ± 1.42	2.56 ± 1.74	0.099	0.436	0.224	0.049

Values are means ± SD.

P* < 0.05 vs pre.*P* < 0.05 vs control.

CBFV, cerebral blood flow velocity; CVRI, CVR index.

function in older patients with type 2 diabetes. Physiological mechanisms underlying these beneficial effects are not clear, but it is plausible to hypothesize that Nordic walking performed in water may have increased blood flow and shear stress to a greater extent and, in turn, improved endothelium-dependent vasodilation and NO bioavailability (26).

Type 2 diabetes is related to an elevated risk of cerebrovascular disease and vascular dementia (39). Arterial stiffness and subsequent loss of the intrinsic buffering capacity increases

hemodynamic pulsatility precipitating cerebrovascular damage and thus cognitive decline (30). A significant negative correlation between HbA_{1c} and cerebral blood flow in all cortical regions might reflect a risk of preclinical oligemia due to altered glycemic control, leading to potential neuronal injury (40). The participants in the present study were elderly patients with type 2 diabetes who exhibited MCI as assessed by MoCA score below 25. Following Nordic walking in water training, average MoCA score was increased approximately 15%. The

TABLE 5. Biochemical variables before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Biochemical Variables	Nonexercising Control		ANW		ANOVA Statistics			Effect Size
	Pre	Post	Pre	Post	Time	Group	Interaction	
WBC (10 ³ ·μL ⁻¹)	7.1 ± 1.6	7.1 ± 1.8	6.8 ± 1.6	6.2 ± 1.3	0.128	0.288	0.140	0.069
RBC (10 ⁶ ·μL ⁻¹)	4.6 ± 0.5	4.5 ± 0.4	4.6 ± 0.4	4.5 ± 0.5	0.119	0.952	0.536	0.012
Hemoglobin (mg·dL ⁻¹)	13.2 ± 1.0	12.9 ± 0.8	13.2 ± 0.5	12.9 ± 0.9	0.019	0.914	0.641	0.007
Hematocrit (%)	40.4 ± 2.8	39.5 ± 2.3	40.2 ± 1.8	39.9 ± 2.7	0.134	0.906	0.454	0.018
Glucose (mg·dL ⁻¹)	147 ± 25	152 ± 40	135 ± 21	117 ± 22*,**	0.033	0.005	0.047	0.129
HbA _{1c} (%)	7.9 ± 0.7	7.6 ± 0.6	7.8 ± 0.7	7.0 ± 0.7*,**	0.001	0.020	0.023	0.156
Insulin (μU·mL ⁻¹)	7.4 ± 2.6	6.8 ± 2.7	6.5 ± 1.9	7.7 ± 2.9	0.484	0.966	0.087	0.101
HOMA-IR (units)	2.94 ± 1.15	3.41 ± 1.56	2.76 ± 1.25	2.17 ± 1.03*,**	0.044	0.015	0.012	0.203
Cholesterol (mg·dL ⁻¹)	164 ± 26	165 ± 22	176 ± 44	171 ± 35	0.536	0.419	0.339	0.030
Triglyceride (mg·dL ⁻¹)	110 ± 47	110 ± 46	110 ± 33	96 ± 31	0.395	0.557	0.378	0.027
HDL (mg·dL ⁻¹)	50 ± 11	52 ± 9	51 ± 11	53 ± 12	0.033	0.735	0.858	0.001
LDL (mg·dL ⁻¹)	98 ± 21	105 ± 24	103 ± 33	96 ± 28	0.928	0.850	0.081	0.101
CRP (mg·L ⁻¹)	1.74 ± 1.1	1.82 ± 1.33	1.77 ± 2.38	1.74 ± 2.24	0.093	0.966	0.797	0.002
IL-1b (pg·mL ⁻¹)	0.52 ± 0.02	0.50 ± 0.02	0.53 ± 0.04	0.53 ± 0.02	0.101	0.012	0.150	0.070
IL-6 (pg·mL ⁻¹)	39.8 ± 2.3	39.6 ± 2.8	39.1 ± 3.8	39.8 ± 2.9	0.750	0.795	0.590	0.010
Nitric oxide (μM)	6.66 ± 2.16	7.17 ± 3.23	6.65 ± 2.13	10.20 ± 3.91*,**	0.002	0.030	0.045	0.141
Adiponectin (pg·mL ⁻¹)	13.8 ± 5.1	12.7 ± 4.7	12.2 ± 5.2	11.1 ± 4.8	0.255	0.286	0.987	0.001
MDA (pg·mL ⁻¹)	1.83 ± 0.57	1.82 ± 0.48	1.91 ± 0.53	1.72 ± 0.38	0.184	0.943	0.273	0.039
TNF-α (pg·mL ⁻¹)	84.7 ± 11.9	86.4 ± 13.8	84.0 ± 13.5	80.8 ± 6.3	0.759	0.387	0.323	0.036
BDNF (pg·mL ⁻¹)	241 ± 16	216 ± 45	225 ± 32	249 ± 26*,**	0.028	0.023	0.002	0.284

Values are means ± SD.

P* < 0.05 vs pre.*P* < 0.05 vs control.

CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; RBC, red blood cell; TNF-α, tumor necrosis factor α; WBC, white blood cell.

TABLE 6. Cognitive function before and after 12 wk of nonexercising control and ANW in elderly participants with type 2 diabetes.

Cognitive Function	Nonexercising Control		ANW		ANOVA Statistics				Kruskal–Wallis Test		
	Pre	Post	Pre	Post	Time	Group	Interaction	Effect Size	df	χ^2	P
MoCA (score)	21.5 ± 2.2	21.2 ± 2.8	22.3 ± 1.5	25.6 ± 1.8 ^{*,**}	0.001	0.001	0.001	0.458	—	—	—
Visuospatial/executive ^a (score)	4 (3–4.5)	4 (2.5–4)	3.5 (3–4)	5 (4–5) ^{**}	—	—	—	—	1	16.4	0.001
Naming ^a (score)	3 (3–3)	3 (3–3)	3 (3–3)	3 (3–3)	—	—	—	—	1	2.0	0.163
Attention ^a (score)	5 (4–5)	5 (4–5.5)	5 (4–5)	5 (4–6)	—	—	—	—	1	1.0	0.311
Language ^a (score)	2 (1–2)	1 (1–2)	2 (1–2)	2 (1–2)	—	—	—	—	1	3.2	0.075
Abstraction ^a (score)	1 (0.5–1)	1 (0–1)	1 (1–1)	2 (1–2) ^{**}	—	—	—	—	1	9.5	0.002
Delayed recall ^a (score)	2 (0.5–3)	3 (2–4)	3 (2–3)	4 (3–4) ^{**}	—	—	—	—	1	4.6	0.033
Orientation ^a (score)	6 (5.5–6)	6 (5–6)	6 (6–6)	6 (6–6) ^{**}	—	—	—	—	1	6.7	0.010
MMSE (score)	26.8 ± 2.6	26.8 ± 1.9	27.7 ± 1.4	28.5 ± 1.0 ^{**}	0.279	0.023	0.212	0.050	—	—	—
Stroop color test (s)	217 ± 45	228 ± 50	213 ± 38	205 ± 28	0.774	0.350	0.066	0.105	—	—	—
Trail Making Test Part B (s)	176 ± 53	176 ± 56	160 ± 64	145 ± 63	0.416	0.217	0.372	0.026	—	—	—

Values are means ± SD or median (interquartile range).

^aKruskal–Wallis test was used to calculate *P* values.

**P* < 0.05 vs pre.

***P* < 0.05 vs control.

improvement in the cognitive function was associated with the corresponding changes in cerebrovascular function as assessed by transcranial Doppler technique. Nordic walking in water improved cognitive performance in type 2 diabetes. This was associated with increased MCA blood velocity possibly mediated by PETCO₂ effects since the cerebrovasculature is very reactive to changes in PETCO₂ and contribute to changes in cerebral perfusion (41). In addition, cerebrovascular resistance was reduced as reflected by higher CVCi during hypercapnia and lower PI in MCA in all conditions. There are a number of possible explanations for the improved cerebrovascular function and the associated improvement in cognitive function induced by Nordic walking in water. As represented in the somatotopic map of primary motor cortex, use of fingers, arms, and shoulders involved in the Nordic walking activates greater areas of the motor cortex in the brain (42). Moreover,

hydrostatic pressure surrounding exercisers translocates blood volume to upper body and increase blood flow and perfusion to the brain (16). The augmented blood flow by a combination of water immersion and aerobic exercise has the potential to enhance shear stress-mediated cerebrovascular adaptation linked with improved brain health (17).

Cerebral circulatory responses to hypocapnia are mediated, at least in part, by endothelial NO (43). We observed increases in functional measures of peripheral endothelial function as well as in biochemical markers of endothelial function in the present study. Arterial stiffness has been implicated in the development of dementia (44), as it is associated with age-related differences in cerebrovascular conductance (27). In addition, patients with diabetes also demonstrate increased PI in the common carotid artery (45). Arterial stiffening, central pulse pressure, and forward wave energy are major contributors to

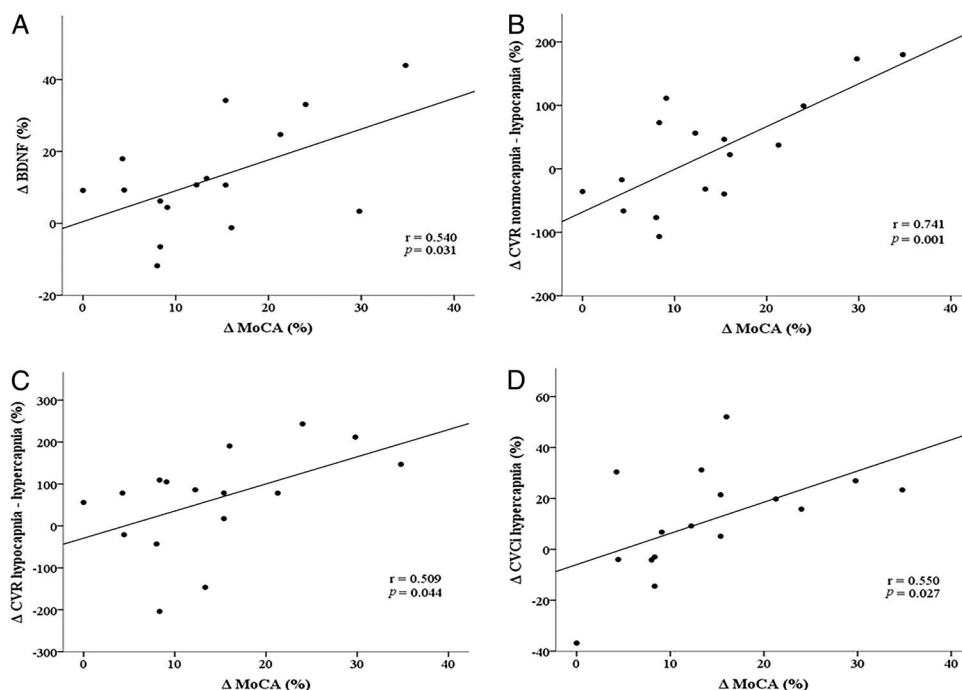


FIGURE 1—Associations of percent changes (pretest and posttest) in MoCA scores with corresponding changes in BDNF (A), CVR normocapnia to hypocapnia (B), CVR hypocapnia to hypercapnia (C), and CVCi during hypercapnia (D).

the transmission of pulsatility into cerebral vessels (46) and could be damaging to the cerebral microvasculature. In the present study, arterial stiffness as measured by baPWV and cerebrovascular stiffness as measured by MCA PI were improved by ANW. Previous findings demonstrated that cerebral reactivity assessed during a cognitive challenge had positive associations with the cognitive domains of response speed, memory, attention, and executive function (47). The improvement in cognitive function was also associated with the corresponding increase in BDNF in the ANW group. BDNF is a neurotrophic factor that plays an important role in protection of neurons from eventual damage, neurogenesis, and neuroplasticity (48). Indeed, the microcirculation in the area of the brain associated with long-term memory is especially vulnerable to pulsatile stress (49). Taken together, ANW may exert cognitive enhancing effects possibly through a variety of underlying mechanisms in patient with type 2 diabetes.

ANW is a whole-body exercise in which both arms and legs are rhythmically moving forward and backward to overcome drag resistance exerted by water. Accordingly, this can be considered one of the multisensory stimulation trainings to emphasize motor coordination between arms and legs. Because exercise is conducted in water, effects of buoyancy minimize excessive joint stress during exercise (50). High thermal conductivity of water lessens heat-related problems that older adults are more prone to (14). As demonstrated in the present study, multitudes of favorable changes carrying clinical and functional significance can be elicited by moderate intensity of exercise that older patient populations can enjoy. There were no participants in the Nordic walking in water group who suffered from injury. Anecdotal evidence during the exercise supervision suggests that all the exercisers enjoyed this

mode of exercise. In fact, the exercise adherence/compliance was extremely high at 100%. This innovative mode of whole-body exercise is a promising exercise that should be evaluated in larger exercise intervention studies involving different patient populations.

There were several limitations in the current study that must be considered. First, the number of participants studied was relatively small, although this sample size is larger than most other exercise training intervention studies involving high-risk patient populations. Second, we did not include control/comparison groups that performed Nordic walking on land as well as walking only or pole swing only groups. Third, because Nordic walking applied to water-based condition is new, there are a number of issues that are unknown. For example, the level of water immersion during Nordic walking needs further investigation. Fourth, because transcranial Doppler flowmeter was used to determine cerebrovascular parameters, a lack of arterial diameter measurement resulting in an inability to derive volumetric blood flow is another limitation of the present study.

CONCLUSIONS

In conclusion, the results of the present study demonstrated that Nordic walking in water exerted beneficial effects on glycemic control, health-related physical fitness, peripheral vascular function, CVR, and cognitive impairment in the elderly patients with type 2 diabetes.

This study was support by the 90th Anniversary of Chulalongkorn University Fund (Ratchadaphiseksomphot Endowment Fund). The authors thank all participants who participated in this study. All authors declare that there are no conflicts of interest. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by American College of Sports Medicine.

REFERENCES

- Messier C, Awad N, Gagnon M. The relationships between atherosclerosis, heart disease, type 2 diabetes and dementia. *Neurol Res.* 2004;26(5):567–72.
- Chen S, Shen Y, Liu YH, et al. Impact of glycemic control on the association of endothelial dysfunction and coronary artery disease in patients with type 2 diabetes mellitus. *Cardiovasc Diabetol.* 2021;20(1):64.
- Sena CM, Pereira AM, Seica R. Endothelial dysfunction—a major mediator of diabetic vascular disease. *Biochim Biophys Acta.* 2013;1832(12):2216–31.
- Chirinos JA, Segers P, Hughes T, Townsend R. Large-artery stiffness in health and disease: JACC state-of-the-art review. *J Am Coll Cardiol.* 2019;74(9):1237–63.
- Skow RJ, Brothers RM, Claassen JAHR, et al. On the use and misuse of cerebral hemodynamics terminology using transcranial Doppler ultrasound: a call for standardization. *Am J Physiol Heart Circ Physiol.* 2022;323(2):H350–7.
- Webb AJ, Simoni M, Mazzucco S, Kuker W, Schulz U, Rothwell PM. Increased cerebral arterial pulsatility in patients with leukoariosis: arterial stiffness enhances transmission of aortic pulsatility. *Stroke.* 2012;43(10):2631–6.
- Nealon RS, Howe PR, Jansen L, Garg M, Wong RH. Impaired cerebrovascular responsiveness and cognitive performance in adults with type 2 diabetes. *J Diabetes Complications.* 2017;31(2):462–7.
- Mazzeo RS, Tanaka H. Exercise prescription for the elderly: current recommendations. *Sports Med.* 2001;31(11):809–18.
- Qiu S, Cai X, Yin H, et al. Exercise training and endothelial function in patients with type 2 diabetes: a meta-analysis. *Cardiovasc Diabetol.* 2018;17(1):64.
- Sharma P, Agarwal M, Tiwari S. Nordic walking: a novel exercise regimen for cardio-metabolic health. A brief review. *J Physiol.* 2017;16:25–30.
- Piotrowicz E, Zieliński T, Bodalski R, et al. Home-based tele-monitored Nordic walking training is well accepted, safe, effective and has high adherence among heart failure patients, including those with cardiovascular implantable electronic devices: a randomised controlled study. *Eur J Prev Cardiol.* 2015;22(11):1368–77.
- Takeshima N, Islam MM, Rogers ME, et al. Effects of Nordic walking compared to conventional walking and band-based resistance exercise on fitness in older adults. *J Sports Sci Med.* 2013;12(3):422–30.
- Rao SM, Binder JR, Hammeke TA, et al. Somatotopic mapping of the human primary motor cortex with functional magnetic resonance imaging. *Neurology.* 1995;45(5):919–24.
- Tanaka H. Swimming exercise: impact of aquatic exercise on cardiovascular health. *Sports Med.* 2009;39(5):377–87.
- Suntraluck S, Tanaka H, Suksom D. The relative efficacy of land-based and water-based exercise training on macro- and microvascular functions in older patients with type 2 diabetes. *J Aging Phys Act.* 2017;25(3):446–52.

16. Pugh CJ, Sprung VS, Ono K, et al. The effect of water immersion during exercise on cerebral blood flow. *Med Sci Sports Exerc.* 2015;47(2): 299–306.
17. Parfitt R, Hensman MY, Lucas SJE. Cerebral blood flow responses to aquatic treadmill exercise. *Med Sci Sports Exerc.* 2017;49(7): 1305–12.
18. Carter HH, Spence AL, Pugh CJ, Ainslie P, Naylor LH, Green DJ. Cardiovascular responses to water immersion in humans: impact on cerebral perfusion. *Am J Physiol Regul Integr Comp Physiol.* 2014; 306(9):R636–40.
19. Bressel E, Louder TJ, Raikes AC, Alphonsa S, Kyvelidou A. Water immersion affects episodic memory and postural control in healthy older adults. *J Geriatr Phys Ther.* 2019;42(4):E1–6.
20. Rosenzweig JL, Weinger K, Poirier-Solomon L, Rushton M. Use of a disease severity index for evaluation of healthcare costs and management of comorbidities of patients with diabetes mellitus. *Am J Manag Care.* 2002;8(11):950–8.
21. Tangwongchai S, Phanasathit M, Chareonboon T, et al. The validity of Thai version of the Montreal Cognitive Assessment (MoCA-T). *Dement Neuropsychol.* 2009;3(2):172.
22. Svensson M. *Nordic Walking*. Champaign (IL): Human Kinetics; 2009. p. 219.
23. Purath J, Buchholz SW, Kark DL. Physical fitness assessment of older adults in the primary care setting. *J Am Acad Nurse Pract.* 2009;21(2):101–7.
24. Rikli RE, Jones CJ. The reliability and validity of a 6-minute walk test as a measure of physical endurance in older adults. *J Aging Phys Act.* 1998;6(4):363–75.
25. Costa EC, Kent DE, Boreskie KF, et al. Acute effect of high-intensity interval versus moderate-intensity continuous exercise on blood pressure and arterial compliance in middle-aged and older hypertensive women with increased arterial stiffness. *J Strength Cond Res.* 2020; 34(5):1307–16.
26. Mitranun W, Deerochanawong C, Tanaka H, Suksom D. Continuous vs interval training on glycemic control and macro- and microvascular reactivity in type 2 diabetic patients. *Scand J Med Sci Sports.* 2014;24(2):e69–76.
27. Jaruchart T, Suwanwala NC, Tanaka H, Suksom D. Arterial stiffness is associated with age-related differences in cerebrovascular conductance. *Exp Gerontol.* 2016;73:59–64.
28. Lau VI, Jaidka A, Wiskar K, et al. Better with ultrasound: transcranial Doppler. *Chest.* 2020;157(1):142–50.
29. Tarumi T, Gonzales MM, Fallow B, et al. Cerebral/peripheral vascular reactivity and neurocognition in middle-age athletes. *Med Sci Sports Exerc.* 2015;47(12):2595–603.
30. Heffernan KS, Augustine JA, Lefferts WK, et al. Arterial stiffness and cerebral hemodynamic pulsatility during cognitive engagement in younger and older adults. *Exp Gerontol.* 2018;101:54–62.
31. Alagiakrishnan K, Zhao N, Mereu L, Senior P, Senthilselvan A. Montreal cognitive assessment is superior to standardized mini-mental status exam in detecting mild cognitive impairment in the middle-aged and elderly patients with type 2 diabetes mellitus. *Biomed Res Int.* 2013;2013:186106.
32. Bajaj JS, Thacker LR, Heuman DM, et al. The stroop smartphone application is a short and valid method to screen for minimal hepatic encephalopathy. *Hepatology.* 2013;58(3):1122–32.
33. Dianatinasab A, Koroni R, Bahramian M, et al. The effects of aerobic, resistance, and combined exercises on the plasma irisin levels, HOMA-IR, and lipid profiles in women with metabolic syndrome: a randomized controlled trial. *J Exerc Sci Fit.* 2020;18(3):168–76.
34. Stringuetta Belik F, Oliveira E Silva VR, Braga GP, et al. Influence of intradialytic aerobic training in cerebral blood flow and cognitive function in patients with chronic kidney disease: a pilot randomized controlled trial. *Nephron.* 2018;140(1):9–17.
35. Grubbs FE. Procedures for detecting outlying observations in samples. *Dent Tech.* 1969;11(1):1–21.
36. Sentinelli F, La Cava V, Serpe R, et al. Positive effects of Nordic walking on anthropometric and metabolic variables in women with type 2 diabetes mellitus. *Sci Sports.* 2015;30(1):25–32.
37. Pippi R, Di Blasio A, Aiello C, et al. Effects of a supervised Nordic walking program on obese adults with and without type 2 diabetes: the C.U.R. I.A.Mo. centre experience. *J Funct Morphol Kinesiol.* 2020;5(3):62.
38. Hauptenthal A, Ruschel C, Hubert M, de Brito Fontana H, Roesler H. Loading forces in shallow water running in two levels of immersion. *J Rehabil Med.* 2010;42(7):664–9.
39. Luchsinger JA, Reitz C, Patel B, Tang MX, Manly JJ, Mayeux R. Relation of diabetes to mild cognitive impairment. *Arch Neurol.* 2007;64(4):570–5.
40. Zlokovic BV. Neurovascular pathways to neurodegeneration in Alzheimer's disease and other disorders. *Nat Rev Neurosci.* 2011; 12(12):723–38.
41. Shoemaker LN, Wilson LC, Lucas SJE, Machado L, Thomas KN, Cotter JD. Swimming-related effects on cerebrovascular and cognitive function. *Physiol Rep.* 2019;7(20):e14247.
42. Roux FE, Niare M, Charni S, Giussani C, Durand JB. Functional architecture of the motor homunculus detected by electrostimulation. *J Physiol.* 2020;598(23):5487–504.
43. Iadecola C. Does nitric oxide mediate the increases in cerebral blood flow elicited by hypercapnia? *Proc Natl Acad Sci U S A.* 1992;89(9):3913–6.
44. Bailey TG, Klein T, Meneses AL, et al. Cerebrovascular function and its association with systemic artery function and stiffness in older adults with and without mild cognitive impairment. *Eur J Appl Physiol.* 2022;122(8):1843–56.
45. Bill O, Mazya MV, Michel P, et al. Intima-media thickness and pulsatility index of common carotid arteries in acute ischaemic stroke patients with diabetes mellitus. *J Clin Med.* 2022;12(1):246.
46. Lefferts WK, DeBlois JP, Augustine JA, Keller AP, Heffernan KS. Age, sex, and the vascular contributors to cerebral pulsatility and pulsatile damping. *J Appl Physiol (1985).* 2020;129(5):1092–101.
47. Cilhoroz BT, DeBlois JP, Lefferts WK, et al. Exploration of cerebral hemodynamic pathways through which large artery function affects neurovascular coupling in young women. *Front Cardiovasc Med.* 2022;9:914439.
48. Budni J, Belletini-Santos T, Mina F, Garcez ML, Zugno AI. The involvement of BDNF, NGF and GDNF in aging and Alzheimer's disease. *Aging Dis.* 2015;6(5):331–41.
49. Conley AC, Karayanidis F, Jolly TAD, Yang MH, Hsieh S. Cerebral arterial pulsatility and global white matter microstructure impact spatial working memory in older adults with and without cardiovascular risk factors. *Front Aging Neurosci.* 2020;12:245.
50. Alkatan M, Baker JR, Machin DR, et al. Improved function and reduced pain after swimming and cycling training in patients with osteoarthritis. *J Rheumatol.* 2016;43(3):666–72.